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Cloudera AI

Hands-On Exercise

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Agenda

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3	Exercise #2 — Data Exploration Using Notebook
4	Exercise #3 — Agent Development
5	Exercise #4 — Bring Your Agent to Your Data
6	More Advanced Topics, Discussions, and Q&A

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Cloudera AI: Build and Run Enterprise AI at scale

Dev

CLOUDERA AI WORKBENCH

Applications and Agents

AI Studios

RAG Studio | Synthetic Data Studio | Fine Tuning Studio | Agent Studio

Notebooks | AMPs

Model training and Fine Tuning

Prod

CLOUDERA AI INFERENCE SERVICE

Autoscaling & High Availability



NVIDIA NIM Microservices

NVIDIA

Monitoring

Enterprise Security & Governance

CLOUDERA AI MODEL REGISTRY

ON PREMISES & CLOUD

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Cloudera AI Hands-On Exercise

Cloudera Platform URL:
https://psedemo1.clouderapartners.click/realms/master/protocol/saml/clients/cdp-sso

Username: workshop-user001 ... workshop-user035
Password: changeme

Workshop Git Repo:
https://github.com/irvanzarkasie/cloudera-ai-air-quality

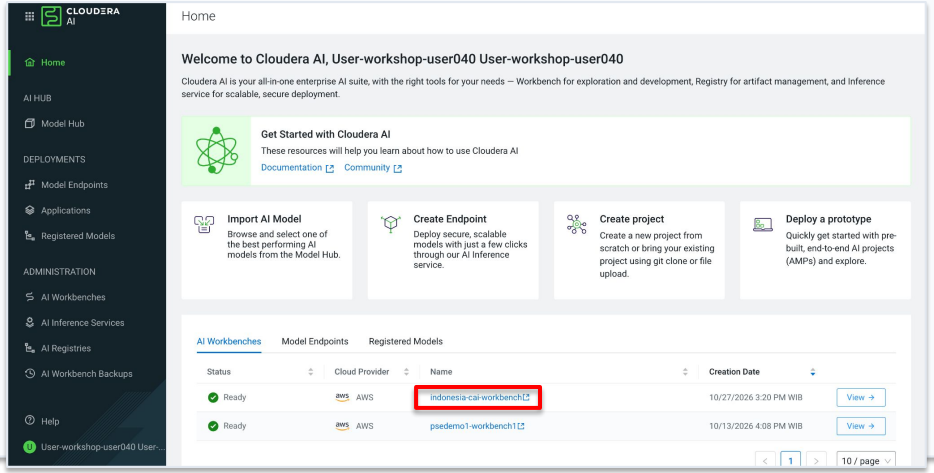
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9cLTVdZVPMDS2ksA

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Preliminary Steps

1. Open the workshop Cloudera Platform URL in your browser:
   https://psedemo1.clouderapartners.click/realms/master/protocol/saml/clients/cdp-sso
2. Sign in with the workshop user account
3. After login, you should land on the Cloudera Platform home page with product tiles
4. From the platform home page, click the Cloudera AI tile to open the Cloudera AI service menu
5. On the Cloudera AI menu, you will see several menus. For our hands-on session, we will be exploring some of the Cloudera AI Workbench features, specifically the **indonesia-cai-workbench** workbench.



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Exercise #1
Cloudera AI Workbench Project

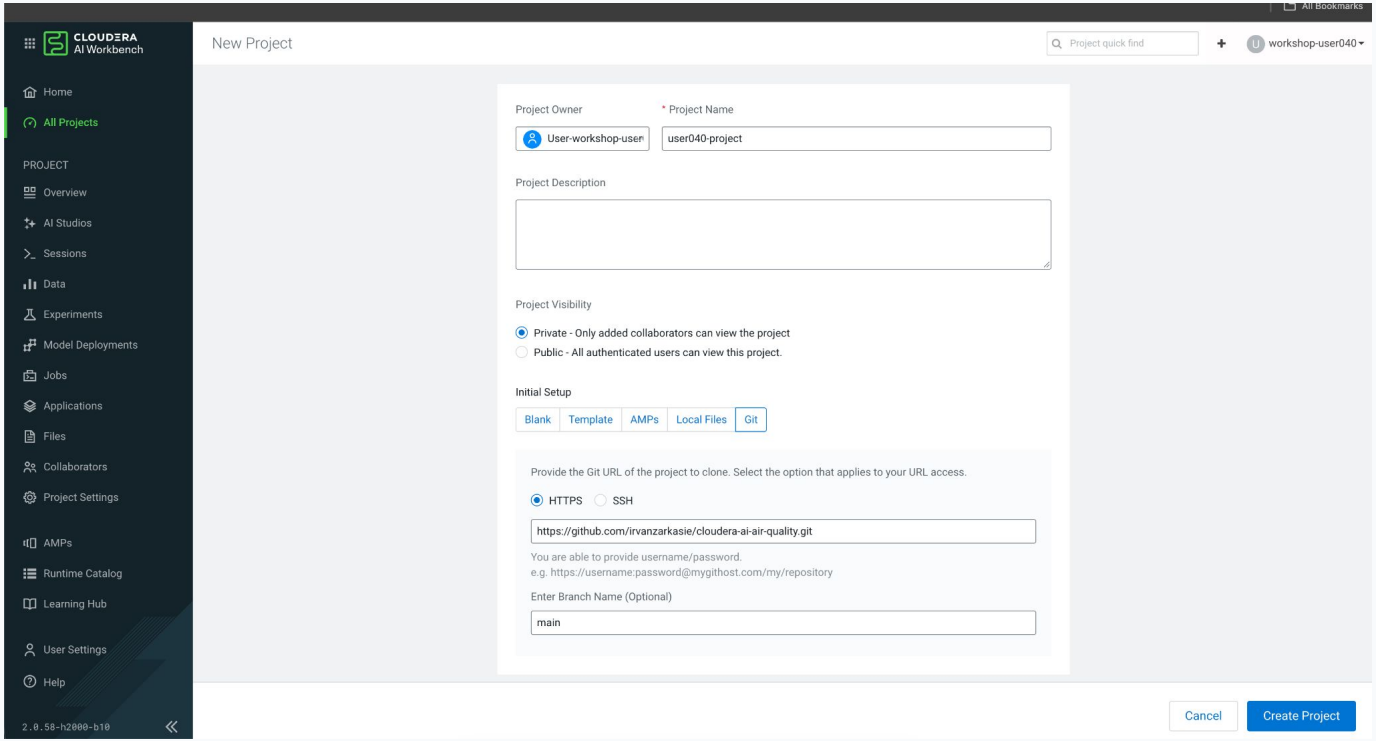
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Exercise #1: Cloudera AI Workbench Project

1. In Cloudera AI Workbench, go to **Projects** and start **New Project**; pick a project name you will recognize.
2. Under Initial setup, choose **Git** so the lab files are cloned from GitHub when the project is created; do not pick Blank, Template, or Local Files for this exercise.
3. Set Protocol to **HTTPS** and paste the repository URL https://github.com/irvanzarkasie/cloudera-ai-air-quality.git and use the **main** branch from the repo.
4. Turn **Advanced Options** on so you can define which session runtimes this project allows; default lists are often Python 3.10 only, which is not what this lab targets.
5. Under **Runtimes**, add one row: **Editor = JupyterLab, Kernel = Python 3.11, Edition = Standard, Version = 2026.08**, then click **Add Runtime** so it appears in the project runtime list.
6. Click **Create Project** and wait for the clone to finish; you should see lab files at the project root (`Jakarta-Air-Snapshot-Lab.ipynb`, `main_v1.py`, `agent_tools/`, `requirements*.txt`, `README.md`).

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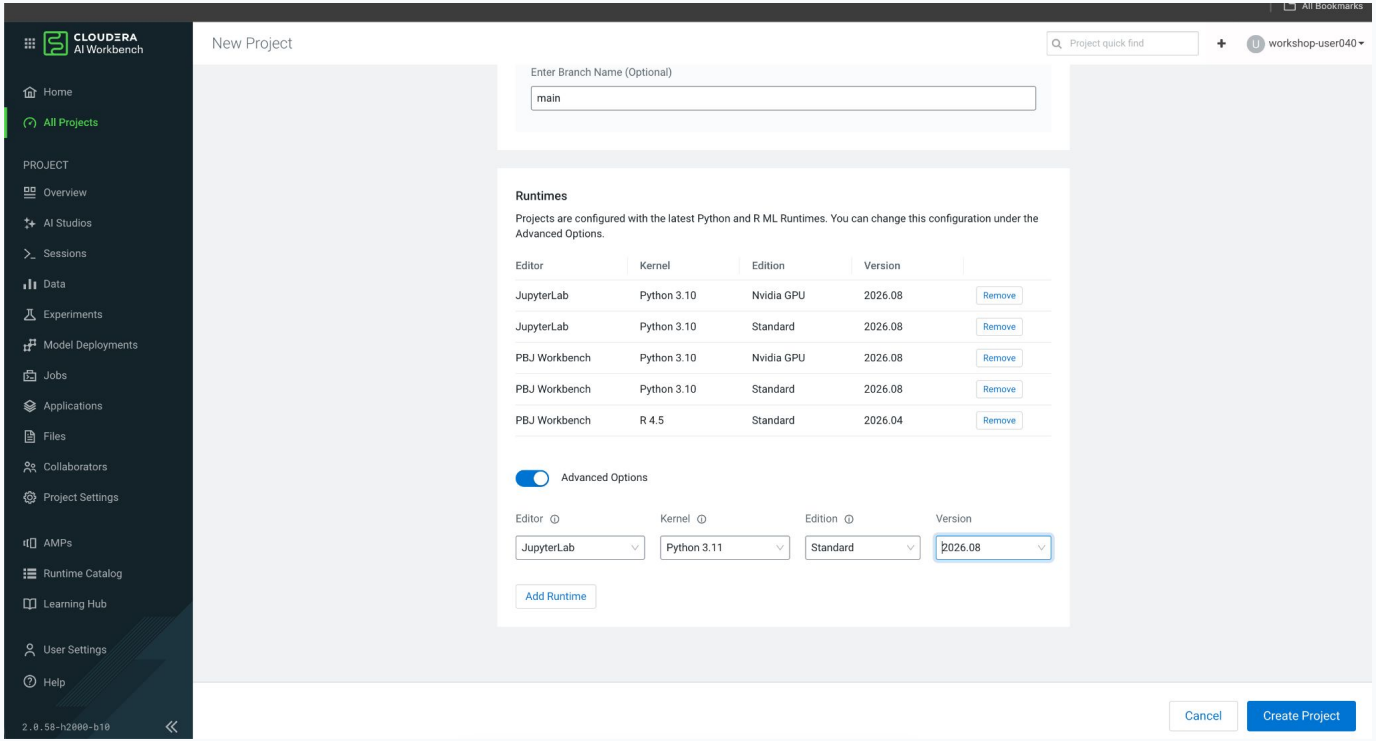
Exercise #1: Cloudera AI Workbench Project



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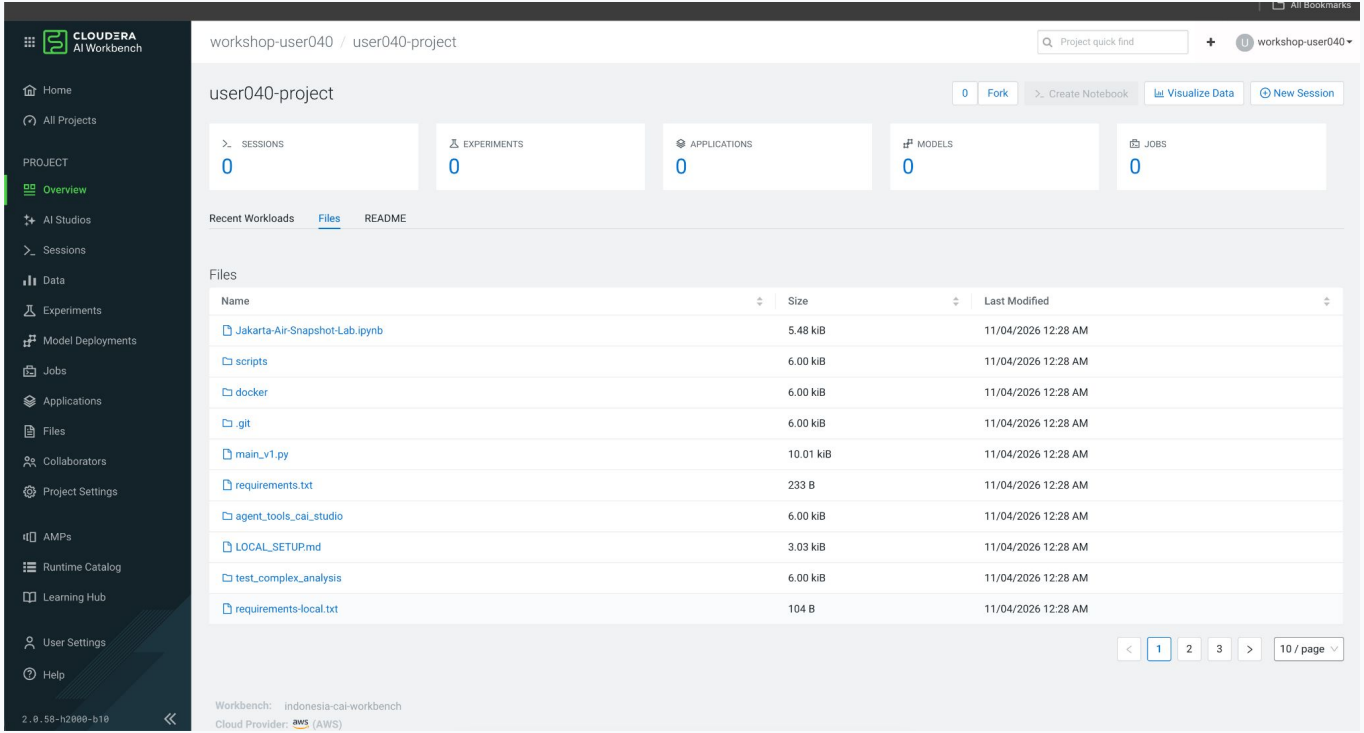
Exercise #1: Cloudera AI Workbench Project



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Exercise #1: Cloudera AI Workbench Project



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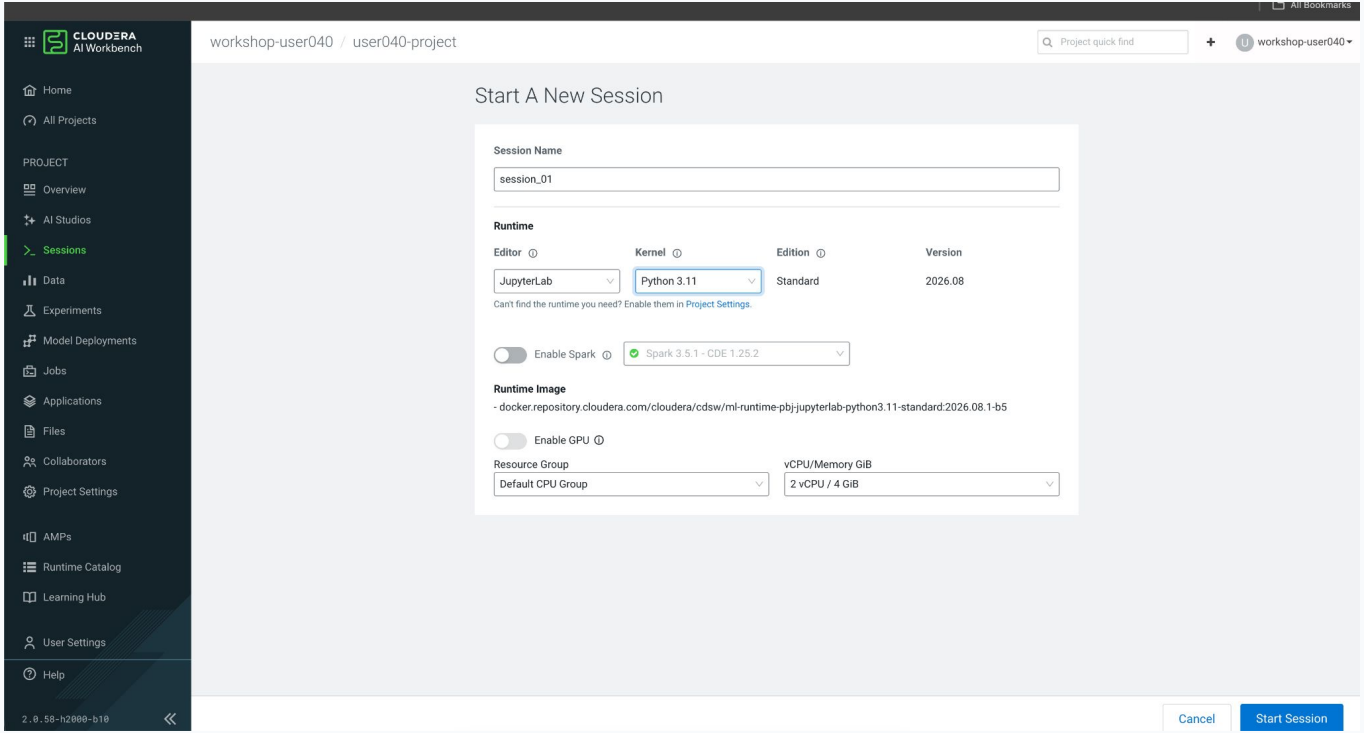
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Exercise #1: Cloudera AI Workbench Project

7. Open **Sessions** (or **New Session** from the project) and launch a session using **JupyterLab**, **Python 3.11**, **Standard**, **2026.08 version.**
8. Open a terminal in that session and confirm you are in the project root (you should see README.md and the notebook when you run ls).
9. In the session terminal, upgrade pip, then install only the minimal requirements:
   > pip install --upgrade pip
   > pip install -r requirements-minimal.txt
   This step should be relatively fast (often about a minute) and installs what you need for Exercise #2: pandas, requests, folium, and related packages for live Udara fetch and the map.
10. When this command completes successfully, in the same terminal, run the full requirements file:
    > pip install -r requirements.txt
11. This pulls in the additional stack and many transitive packages; downloads may finish quickly, but the install phase may take some time.
12. You can leave this command running while you work through the notebook in Exercise #2.

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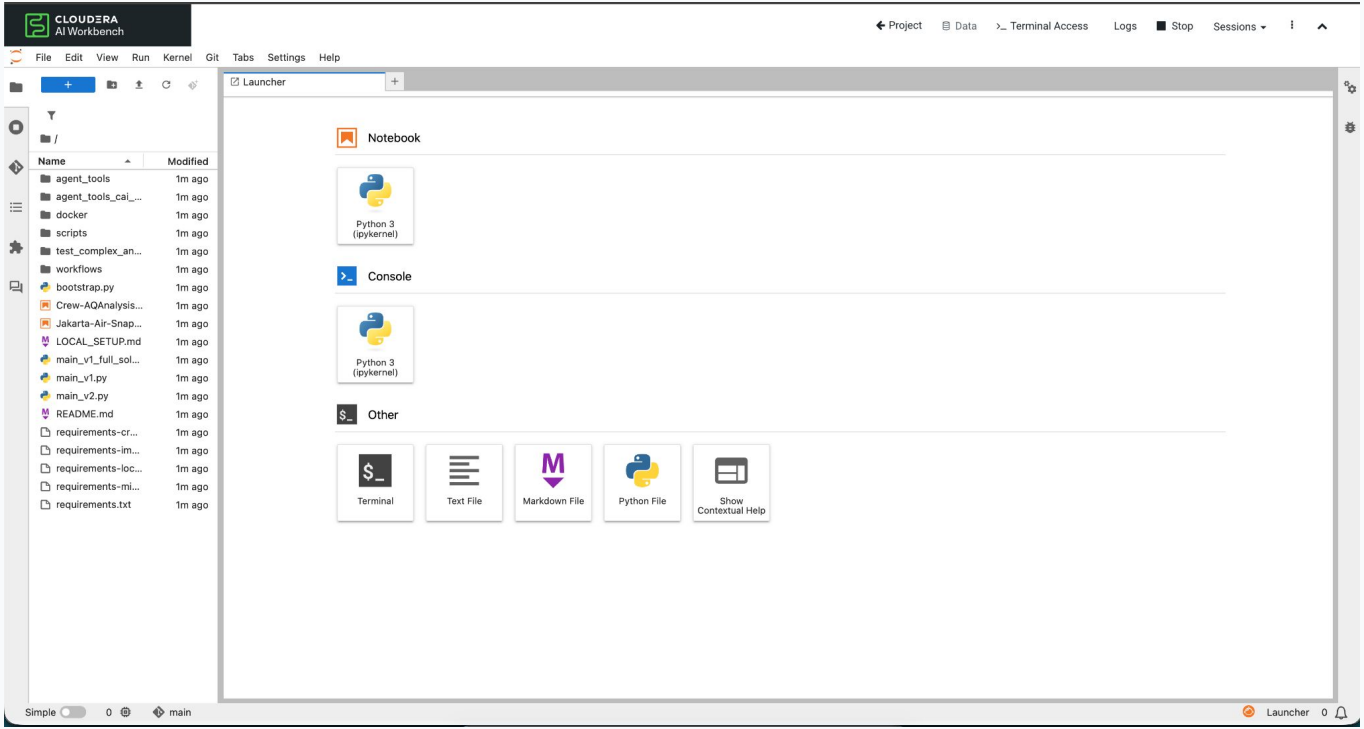
Exercise #1: Cloudera AI Workbench Project



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Exercise #1: Cloudera AI Workbench Project



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Exercise #2
Data Exploration Using Notebook

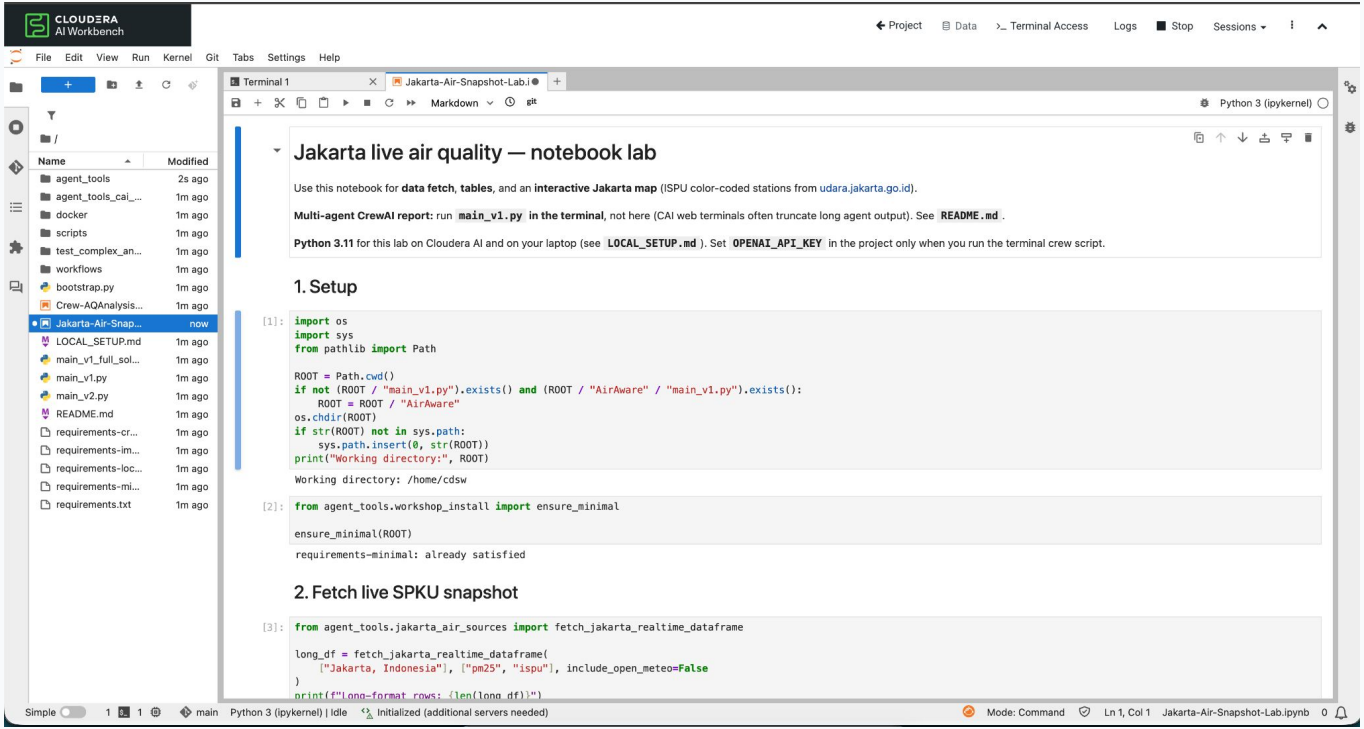
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Exercise #2: Data Exploration Using Notebook

1. In your JupyterLab session, open **Jakarta-Air-Snapshot-Lab.ipynb** from the project root and confirm the kernel is tied to your Python 3.11 session.
2. Try to inspect and explore the notebook. This notebook is intentionally limited to data fetch, tables, and visualization.
3. Run the first code cells to set the working directory to the project root (where main_v1.py lives) and to ensure minimal Python packages are available via agent_tools/workshop_install.py.
4. Execute the cells that call agent_tools/jakarta_air_sources.py and agent_tools/jakarta_viz.py to pull the current SPKU snapshot from udara.jakarta.go.id and build a station-level table.
5. Review the printed number of stations, sample rows, and especially the Udara update time; the lab uses a live snapshot.
6. Parameters such as pm25 and ispu come from the official portal embed; note how kecamatan appears next to station names in the table.
7. Run the map cell to render an interactive Folium map with stations colored by ISPU band (greener = better air quality, redder/purpler = worse).
8. When the map displays correctly, you have completed the core outcome of Exercise #2; next you will wire the same data sources into agents in Exercise #3.

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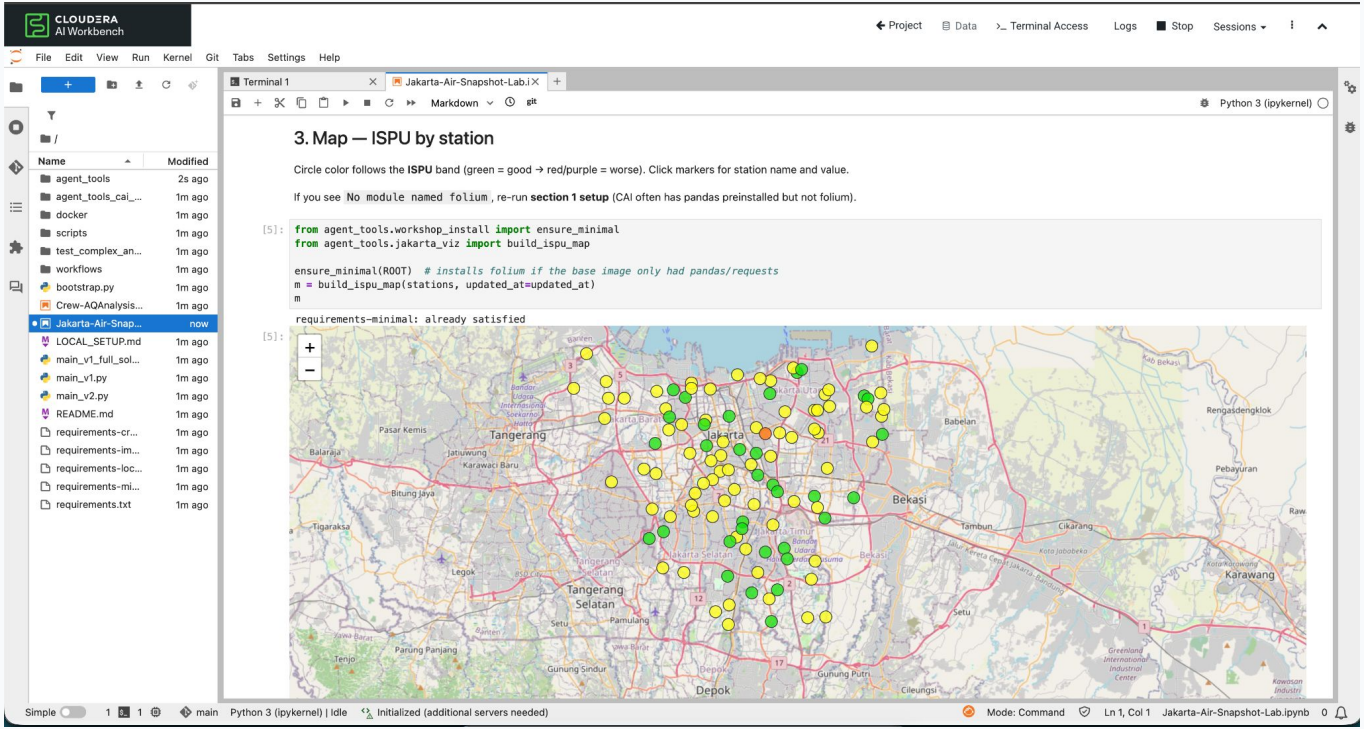
Exercise #2: Data Exploration Using Notebook



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Exercise #2: Data Exploration Using Notebook



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Exercise #3
Agent Development

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Exercise #3: Agent Development – Part A

1. On this exercise, we will build an agent that can answer “What is Jakarta air quality right now?” by chaining tools (parse user text, fetch weather, fetch Udara snapshot) and an analyst LLM that writes a short markdown report.
2. Before Exercise #3 can be started, confirm pip install -r requirements.txt has finished; then in the project terminal run python main_v1.py --dry-run.
3. Read the printed snapshot text and weather summary; if dry-run fails, fix network or minimal deps before involving the LLM.
4. In **main_v1.py**, you define **CrewAI Agents and Tasks:** an input parser, a weather specialist, an air-quality retriever, and an analyst with an LLM.
5. Each tool-using agent is given the matching class from agent_tools/ (InputParserTool, AirQualityAnalysisTool, CurrentWeatherTool); tasks are ordered so later steps receive context from earlier ones.
6. Set **OPENAI_API_KEY**, then run the crew with a natural-language question, for example asking for Jakarta pm25 and ispu and the worst and best stations today.

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Exercise #3: Agent Development – Agent Definitions

No	Agent Role	Agent Goal	Backstory
1	Input Data Parser	Extract locations and air quality parameters for a live snapshot report.	Parses natural language into structured fields for Jakarta live air quality. Ignores historical date ranges if the user mentions them.
2	Current Weather Specialist	Retrieve current weather for the same locations as the air quality snapshot.	Summarizes present meteorological conditions that may affect pollution dispersion.
3	Air Quality Data Retriever	Fetch the latest official Jakarta SPKU/LCS readings from udara.jakarta.go.id using air_quality_tool tools.	Specialized in live DKI DLH air quality snapshots.
4	Air Quality Analyst	Produce a concise snapshot report from live station readings and current weather — no historical trend analysis.	Experienced in interpreting ISPU/PM levels across Jakarta monitoring stations.

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Exercise #3: Agent Development – Task Definitions (page 1 from 2)

No	Task Description	Assigned Agent	Expected Output
1	Parse the user input: {user_input}. Use InputParserTool to extract locations and aq_parameters only. Do not require or validate start/end dates.	input_parser_agent	Validated locations, aq_parameters, mode=live_snapshot, data_source=udara.jakarta.go.id.
2	From parsed locations in context, call CurrentWeatherTool for each location (or once for Jakarta). Summarize current temperature, humidity, wind, and precipitation.	weather_data_integrator	Current weather summary per location with observation time.
3	Call air_quality_tool with locations and aq_parameters from context. Pass bounding_boxes=[]; dates are optional and ignored. Return the live snapshot text from udara.jakarta.go.id (includes ISPU extremes summary).	air_quality_retriever	Air quality snapshot text with observation date, ISPU extremes, and sample rows.

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Exercise #3: Agent Development – Task Definitions (page 2 from 2)

No	Task Description	Assigned Agent	Expected Output
4	User request: {user_input} Write a live snapshot report (not historical trends). Use the ISPU extremes block and weather context from prior tasks. Highlight the worst and best ISPU stations with station names and values. Do not enumerate every station — at most 5 highest and 5 lowest ISPU unless the user asked for a full table. State observation time, brief ISPU category explanation, weather/dispersion note, Summary and Conclusion. Source: udara.jakarta.go.id. Plain markdown, complete sentences.	air_quality_analyst	Complete markdown report (not truncated mid-sentence) with worst/best ISPU stations.

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Exercise #3: Agent Development – Agent Tooling Assignment

No	Agent Role	Tools Used
1	Input Data Parser	input_parser_tool
2	Current Weather Specialist	weather_tool
3	Air Quality Data Retriever	air_quality_tool
4	Air Quality Analyst	N/A

air_quality_tool will require the following dependencies (requirements.txt)

```
# https://pip.pypa.io/en/stable/reference/requirements-file-format/
pydantic
boto3==1.38.17
pandas==2.2.3
```

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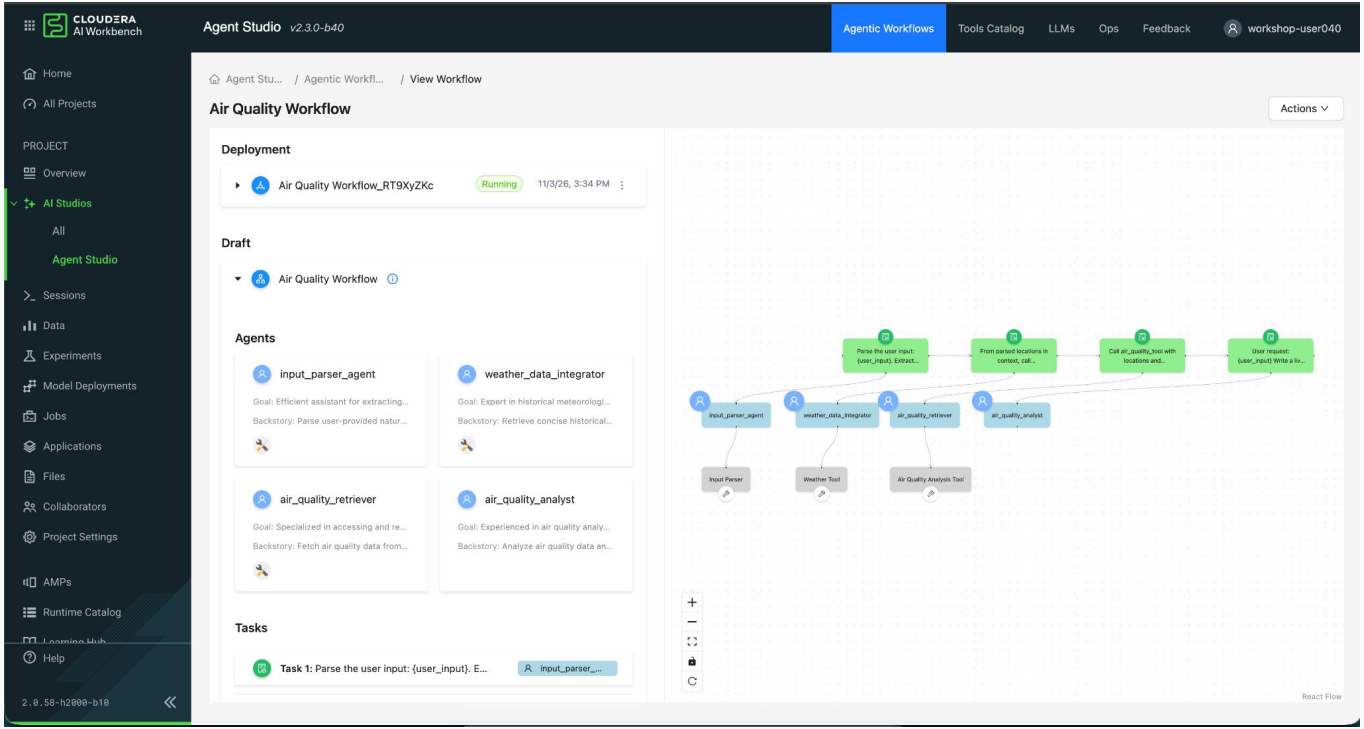
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Exercise #3: Agent Development – Part B

1. In **Cloudera AI Workbench**, open your lab project. In the left project menu, click **AI Studios** to expand it; you should see entries such as All and Agent Studio under that section.
2. If you have never started **Agent Studio** in this project, click **Launch** → on that card. If you already launched it earlier (including in a prior exercise), the card may show **Open** → instead—use **Open** to return to the same Agent Studio instance.
3. If Running Studios lists Agent Studio with status Success, you can also use Open from that row; wait until status is Success before continuing. If launch fails, ask the facilitator before retrying.
4. After Launch or Open, your browser should show **Agent Studio** (top bar) with tabs such as **Agentic Workflows**, **Tools Catalog**, **LLMs**, **Ops**, and **Feedback**.
5. In the Agent Studio menu, create or open a workflow where you register the workshop tools from agent_tools_cai_studio/ (input parser, air quality analysis, weather).
6. Connect agents and tool steps so execution order matches main_v1.py: parse the user message, fetch current weather, fetch the live AQ snapshot, then let an analyst-style agent produce markdown with worst/best ISPU and observation time.
7. Test with the same style of prompt you used in the terminal; the behavior should be consistent with the Step 3 Part A, not a different analytics story.

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Exercise #3: Agent Development



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Exercise #4
Bring Your Agent to Your Data

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Exercise #4: Bring Your Agent to Your Data

1. In Exercises #2–#3 you used live web data (Udara + weather); in Exercise #4 you connect an Agent Studio workflow to warehouse data in id_workshop.jakarta_air_daily using the prebuilt template “Talk to your CDW Database.”
2. Find the workflow template titled exactly “Talk to your CDW Database” (not the Kerberos variant). You will use this template to talk to the shared id_workshop database.
3. Click the “Talk to your CDW Database” template card once to select or open it.
4. Click the Actions control for that template.
5. From the menu, choose Create Workflow from Template so Agent Studio creates a new editable workflow in your project instead of modifying the shared template master.
6. You should enter the Edit Workflow wizard with steps such as Add Agents → Add Tasks → Configure → Test → Deploy; you may click through Add Agents and Add Tasks with defaults unless the facilitator asks for changes—the critical attendee work is under Configure

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Exercise #4: Bring Your Agent to Your Data

7. In the wizard, go to step 3. Configure (highlighted when active). On the left you will see model settings and Tools and MCPs; expand the tool attached to the SQL Execution Agent—CDW Hive Database Read Tool.
8. Fill in each field below carefully; values must match the workshop environment and your login.

Field	**What to enter**
hive_cai_data_connection_name	psedemo1-impala-1
workload_user	Your current workshop username
workload_pass	Your workshop account password (for standard workshop users this is usually **changeme**)
default_database	id_workshop
cdsw_api_v2_key	A personal API key generated from Workbench User Settings (see next slide)

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Exercise #4: Bring Your Agent to Your Data – Generating CDSW API Key

- The cdsw_api_v2_key field is required so Agent Studio can call Workbench APIs securely.
- In **Cloudera AI Workbench**, open **User Settings** from your user menu (top-right avatar or name—for example workshop-user040), **not Project Settings**.
- Find the section for **API keys / CDSW API v2 key** (wording may vary slightly by version), click **Generate** or **Create key**, copy the key once, and paste it into **cdsw_api_v2_key** in the workflow Configure step.

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Exercise #4: Bring Your Agent to Your Data

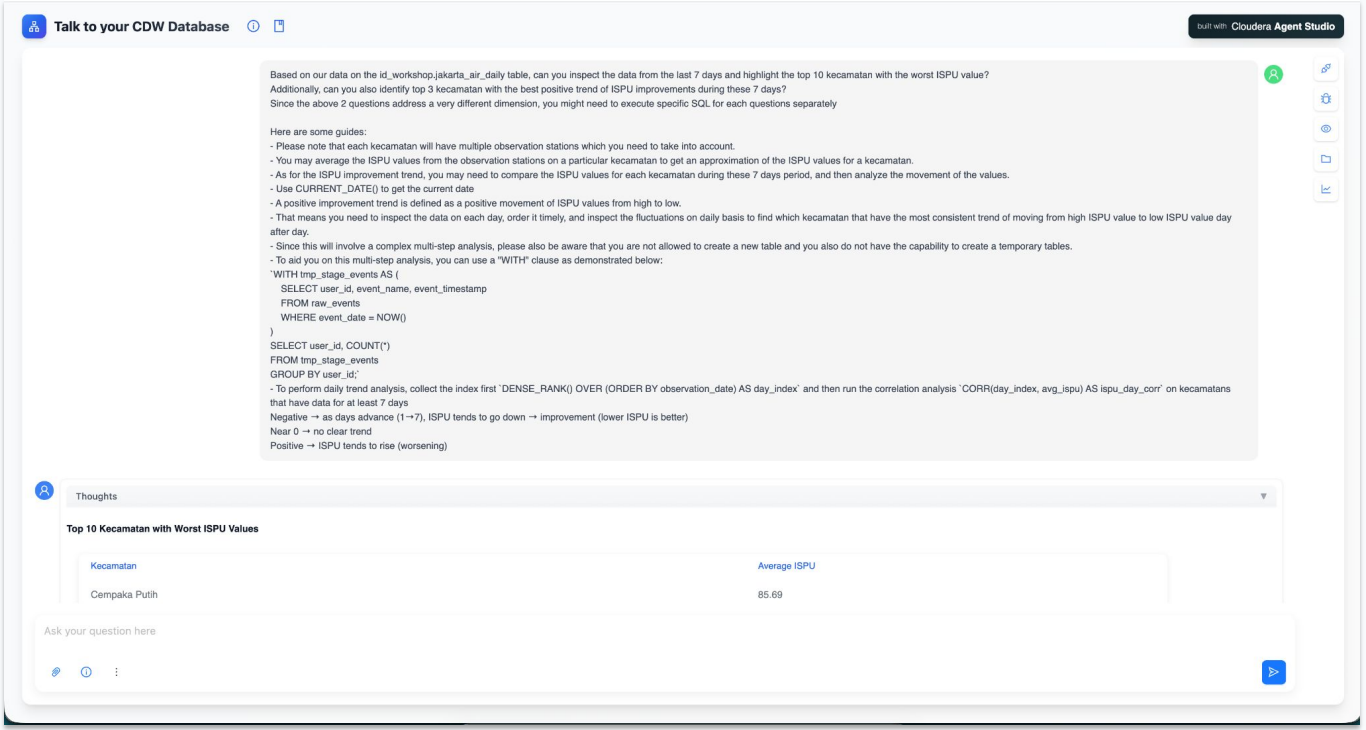
9. Use the following prompt to test the agent:

```
Based on our data on the id_workshop.jakarta_air_daily table, can you inspect the data from the last 7 days and highlight the top 10 kecamatan with the worst ISPU value?
Additionally, can you also identify top 3 kecamatan with the best positive trend of ISPU improvements during these 7 days?
Since the above 2 questions address a very different dimension, you might need to execute specific SQL for each questions separately

Here are some guides:
- Please note that each kecamatan will have multiple observation stations which you need to take into account.
- You may average the ISPU values from the observation stations on a particular kecamatan to get an approximation of the ISPU values for a kecamatan.
- As for the ISPU improvement trend, you may need to compare the ISPU values for each kecamatan during these 7 days period, and then analyze the movement of the values.
- Use CURRENT_DATE() to get the current date
- A positive improvement trend is defined as a positive movement of ISPU values from high to low.
- That means you need to inspect the data on each day, order it timely, and inspect the fluctuations on daily basis to find which kecamatan that have the most consistent trend of moving from high ISPU value to low ISPU value day after day.
- Since this will involve a complex multi-step analysis, please also be aware that you are not allowed to create a new table and you also do not have the capability to create a temporary tables.
- To aid you on this multi-step analysis, you can use a "WITH" clause as demonstrated below:
`WITH tmp_stage_events AS (
    SELECT user_id, event_name, event_timestamp
    FROM raw_events
    WHERE event_date = NOW()
)
SELECT user_id, COUNT(*)
FROM tmp_stage_events
GROUP BY user_id;`
- To perform daily trend analysis, collect the index first `DENSE_RANK() OVER (ORDER BY observation_date) AS day_index` and then run the correlation analysis `CORR(day_index, avg_ispu) AS ispu_day_corr` on kecamatans that have data for at least 7 days
Negative → as days advance (1→7), ISPU tends to go down → improvement (lower ISPU is better)
Near 0 → no clear trend
Positive → ISPU tends to rise (worsening)
```

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Exercise #4: Bring Your Agent to Your Data



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More Advanced Topics, Discussions, and Q&A

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Thank You

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